

Homework 2 Solutions:

Shake It Off (The Budget Concerns):
Assessing the Economic Impact of Hosting Taylor Swift's Eras Tour

Quantitative Methods II - Winter 2025

February 17, 2025

Acknowledgments

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3.1 Preliminaries

Q1: How many observations and variables are in the working dataset at this point?

A1: There are 14,416 observations and 42 variables

Q2: According to the U.S. Department of Agriculture's Economic Research Service, the Rural-Urban Continuum Codes (RUCC) classify U.S. counties into nine categories based on metropolitan status, population size, and degree of urbanization. Code 1 corresponds to counties in metropolitan areas with a population of 1 million or more. How many such counties are represented in the working dataset?

A2: 424 distinct counties

Q3: If each row in the working dataset represents a county-level observation, and there are only a few hundred counties in U.S. metropolitan areas with populations of 1 million or more, why does the working _data contain thousands of rows?

A3: It contains thousands of rows for each year and per industry. Filtered to show counties with only 1 million or more population.

Q4: What do the variables 2001, 2002, up to 2023 represent in the dataset at this point?

A4: They represent the GDP in thousands of dollars for that counties industry for that year.

3.4.4 Transforming the Outcome Variables

Q5: In this section, we applied two transformations to the outcome variables: first, we converted GDP to per capita by dividing it by population, and second, we transformed the per capita values by taking their logarithm. **Q5a:** Why was it important to adjust for population size when analyzing GDP? **Q5b:** Based on the histograms you generated, what issue in the data prompted the need for a log transformation? **Q5c:** Did the log transformation resolve the issue you observed in the initial histogram of GDP per capita? Briefly explain your reasoning.

A5a: We adjust to per capita measures as counties or industries with larger populations naturally would be over representing their GDP. Thereby allowing us to compare countries apple to apples.

A5b: In the initial histogram there was a massive outlier which skews the histogram extremely right. Running a log transform allows us to curb the data set in a way that accounts for outliers and shortens the tail.

A5c: The log transformation does resolve the issue found in the initial histogram. The outliers have been curtailed. By converting GDP to log we can convert changes later in GDP to percentage changes, capturing effects more clearly.

3.5 Analyzing Time Trends in Arts, Entertainment, and Recreation Industry (NAICS 71) Annual Log GDP per Capita

Q6: Examine the plot you generated. How does the trend in log GDP per capita for counties that hosted the Eras Tour compare to those that did not? Specifically, how do the slopes of these trends differ?

A6: Both the hosted and not hosted groups trend upward. However, the hosted groups trend upwards more sharply overtime. With the hosted groups beginning at a higher log of GDP than unhosted groups. Hosted groups began at 0.43 log of GDP while unhosted began much lower at ~0.10 log of GDP.

Q7: Why might this difference in pre-existing trends between the two groups create a problem when estimating the effect of hosting the Eras Tour on GDP?

A7: There is a causal question that since counties which hosted the Eras Tour already had greater GDP, their growth could be contributed to the compounding effects prior to the Eras tour. Thereby posing an omitted variable bias and the assumption of endogeneity could be questioned. As growth of the county may not be able to be easily disentangled from the Eras Tour effect. There are also demographic differences could already be contributing to the GDP difference and rate difference. Further pushing the need to add these variables as controls.

3.6 Visualizing Demographic Differences Across Counties that Hosted the Eras Tour and Those That Did Not

Q8: Examine the bar plots you generated. How do the demographic and socioeconomic characteristics of counties that hosted the Eras Tour compare to those that did not? Specifically, which characteristics show notable differences between the two groups?

A8: The notable differences are level of education. Counties which hosted the Eras Tour, had a higher proportion of the population which had a bachelor's degree or higher, 29% in hosted counties and 21% in nonhost counties. Counties that hosted the Eras Tour also experienced greater negative migration, with decreases of about 4000 residents in 2023, versus increases in nonhost counties of around 1000.

Q9: Why might these differences in demographic and socioeconomic characteristics create a problem when estimating the effect of hosting the Eras Tour on GDP?

A9: Due to stark differences in education level and previous economic literature showing the causal relationship between education level and GDP. These demographic differences pose an omitted variable bias if not controlled for in the regression. As if it is unaccounted for the effect of a more highly educated population may be falsely confounded with the Eras Tour.

3.7 Multiple Regression Analyses

Q10: Look at the coefficient for Eras Tour Host in Model 5 under NAICS 71. What does the estimate tell you about how hosting the **Eras Tour affects county-level GDP** per capita in the **Arts, Entertainment, and Recreation industry**? Is the coefficient statistically significant?

A10: Looking at model 5, the correlation coefficient is 0.016 for the Eras Tour. The relationship is positive at the 95% confidence interval. Which demonstrates on average that in counties where the Eras Tour was hosted, it boosted GDP of the Arts, Entertainment, and Recreation (AER) industry by 1.62%. This shows that while there is a positive correlational relationship with the Eras Tour, its magnitude is quite small.

Q11: How does controlling for GDP per capita in 2022 in Model 5 help address the issue raised in **Q7** regarding pre-existing trends? Why is this step important in reducing omitted variable bias?

A11: By controlling for GDP per capita we are able to isolate the Eras Tour effect, by placing it as a control variable in our final model. This reduces omitted variable bias of previous GDP as a contributing factor. Increasing our model fit, and more importantly isolating the positive effect of the Eras Tour on the AER industry.

Q12: How do the changes in the coefficient estimates for eras tour host across the models address the issue raised in **Q9** regarding omitted variable bias from demographic and socioeconomic differences?

A12: By increasingly controlling for demographic, socioeconomic factors, and GDP we are systematically rooting out the omitted variable bias from our regression. The Eras Tour goes from being statistically significant (at a 95% confidence interval) and its magnitude being 0.5% in the simple bivariate regression, to 1.62%.

Q13: Examine the regression results for the other industries (NAICS 72, NAICS 54, and NAICS 11). Are the results consistent with the hypothesis that the Eras Tour had a greater economic impact on industries most likely to benefit from large events? Provide evidence from the models to support your response.

Figure 1

Industries	Arts, Entertainment, Recreation	Agriculture, Forestry Fishing, & Hunting	Professional, Scientific, & Technical Services	Accommodation & Food Services
Before controls	0.563***	-0.282***	1.101***	0.518***
After controls	0.016**	-0.016	-0.0002	0.006

Figure 1: Shows the before and after controls correlation coefficient of the Eras Tour. Showing the decreasing magnitude of the Eras Tour once controlled for Demographic, Socioeconomic, and GDP factors.

Note: *p<0.1; **p<0.05; ***p<0.01

A13: This effect is even more pronounced when looking at the other industries. As more control variables are added, decreasing omitted variable bias, the effect goes from being statistically significant to be statistically insignificant. Showcasing no correlational relationship of the Eras Tour and GDP. This lends credence to the idea that the Eras Tour does have a greater impact on Industries traditionally thought to benefit from large events like, the AER industry. Albeit with a small magnitude of only increases of on average 1.62% of GDP in counties which hosted the Eras Tour.

Q14: Based on your regression analysis for the Arts, Entertainment, and Recreation industry (NAICS 71), calculate the implied increase in GDP for counties that hosted the Eras Tour. Use the provided hints and data to guide your calculations.

A14: Given the average effect of 1.62% on GDP, and the average GDP for NAICS 71 counties is 4.1 billion. A 1.62% increase in 4.1 billion is a \$66,884,038 increase. Divided by the average population (2.3 million) to produce dollar earned per capita, we show that the average effect of the Eras Tour assuming average GDP of NAICS 71 counties, with a 95% confidence interval, is \$29.08 per person.

Q15: Now, based on your calculation in **Q14**, what would you report to the mayor regarding the potential economic impact of hosting a Taylor Swift concert in San Diego?

1. Evidence of economic impact: Summarize the key findings from the regression analysis, including the size and statistical significance of the effect.

The potential economic impact of hosting the Eras Tour, is positive on the AER industry. However, the effect is relatively small, only increasing GDP on average by 1.62%, or \$29.08 per person in a county of 2.3 million people. These findings are statistically significant at a 95% confidence interval.

2. Estimated economic benefit: Report the dollar increase in county-level GDP per capita and total GDP implied by the model.

The Dollar increase in the county-level GDP per capita, is \$29.08 per person in a county of 2.3 Million people. Its total estimated increase in GDP for AER industry (NAICS 71), is \$29.08 per person in a county of 2.3 million people.

3. Strength of the argument: Explain how controlling for education, poverty, migration, and prior GDP helps provide a more credible estimate of the impact, reducing concerns about omitted variable bias.

Since the counties which hosted the Eras Tour are socioeconomic, and economically different to counties which did not host, we ran into a possible omitted variable bias. For example, counties which did host typically had a more post-secondary school educated population. With previous research showing that higher educated populations typically generate higher GDP, not including this control variable would introduce omitted variable bias. By controlling for demographic variables and pre-Eras Tour GDP in our final regression model, we are able to isolate the true effect of the Eras Tour on the GDP of counties which hosted.

4. Relevance to San Diego: Discuss how these findings support the idea that a major event could boost the local economy and provide a rationale for investing in hosting the concert mindful of the cost.

These findings show that on average, for the AER industry, there is a statistically significant and measurable increase in GDP. While the impact is relatively small, an average of 1.62% increase, we are able to compare if the associated costs and investments to properly host the Eras Tour would make sense from a cost benefit standpoint for the city of San Diego. This analysis has given policy makers an important data point to conduct a comprehensive analysis on the opportunity costs of hosting. If there are other vehicles to increase GDP which have spill overs for other industries outside of AER, or ones which have a higher confidence interval in the relationship, policy makers have a clear-cut choice to make in terms of economics. However, there may be an argument that the cultural significance of the Tour, and the resulting impact on the AER industry through network effects may outweigh the economic costs of hosting.

Appendix:

Figure 1

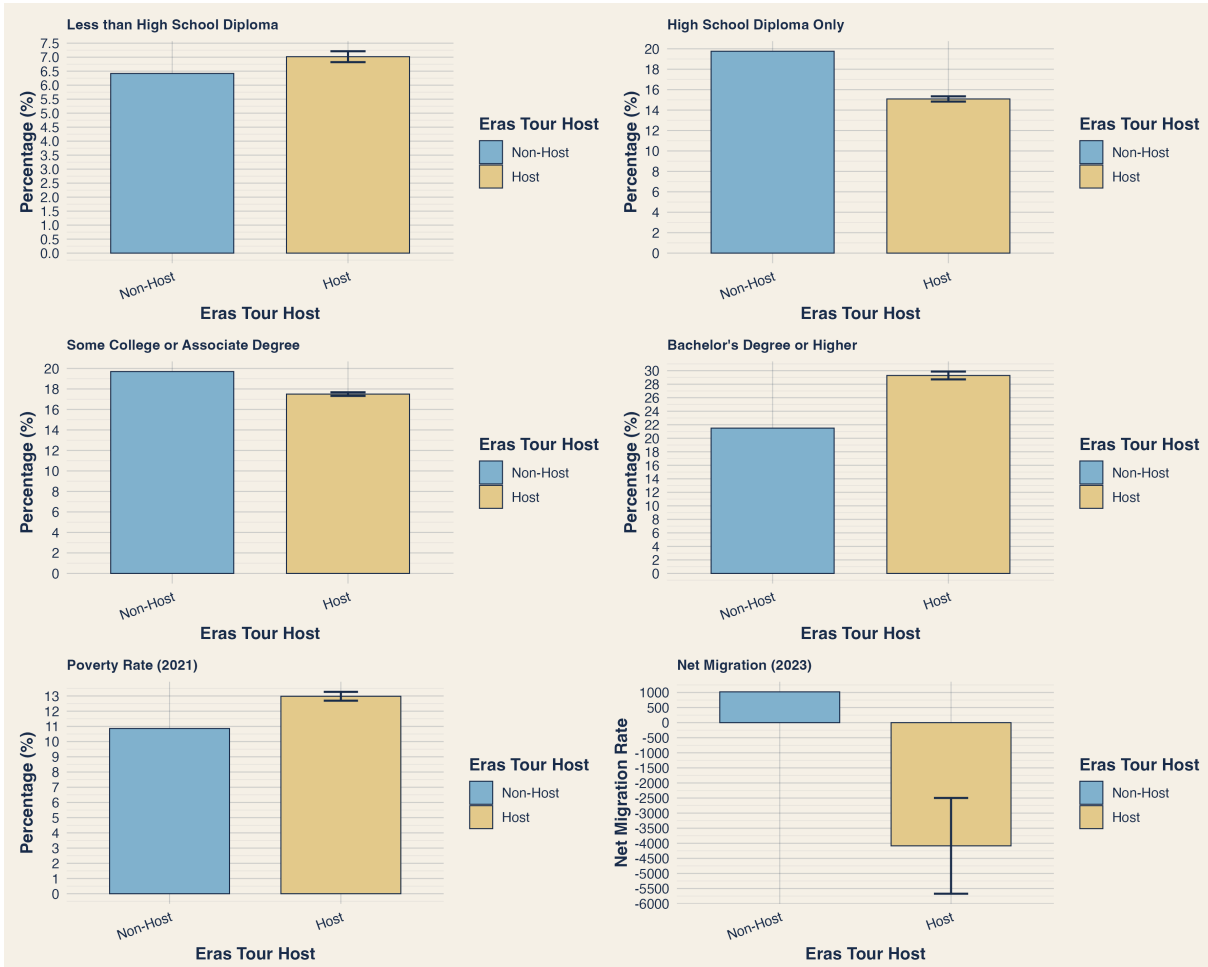


Figure 4 compares demographic and socioeconomic averages between Eras Tour host and non-host counties, with treated-group confidence intervals shown for each metric.

Figure 2

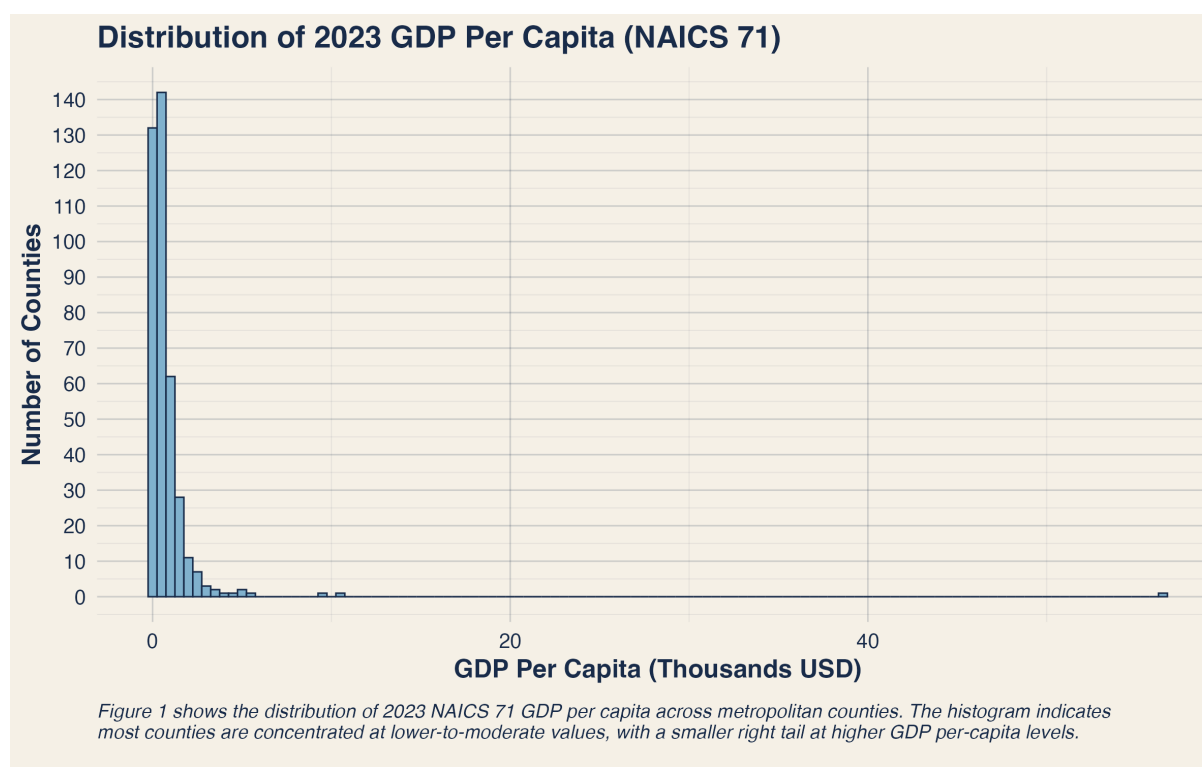


Figure 3

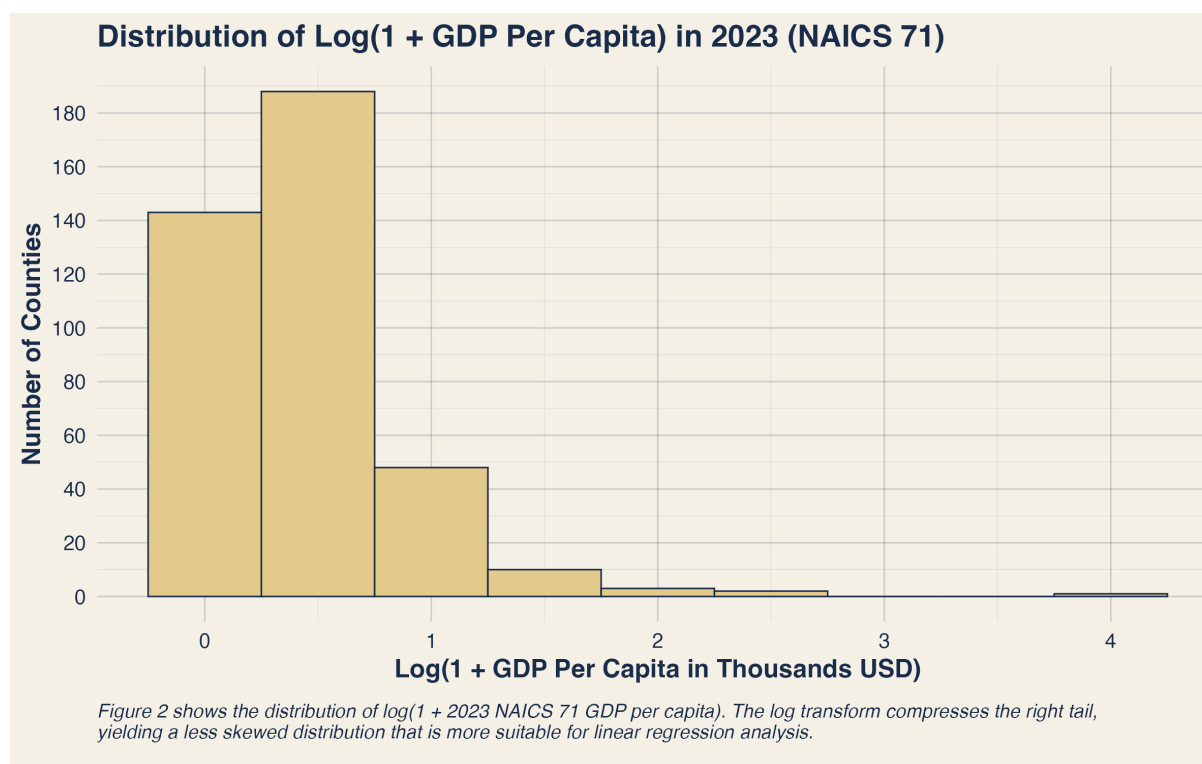


Figure 4

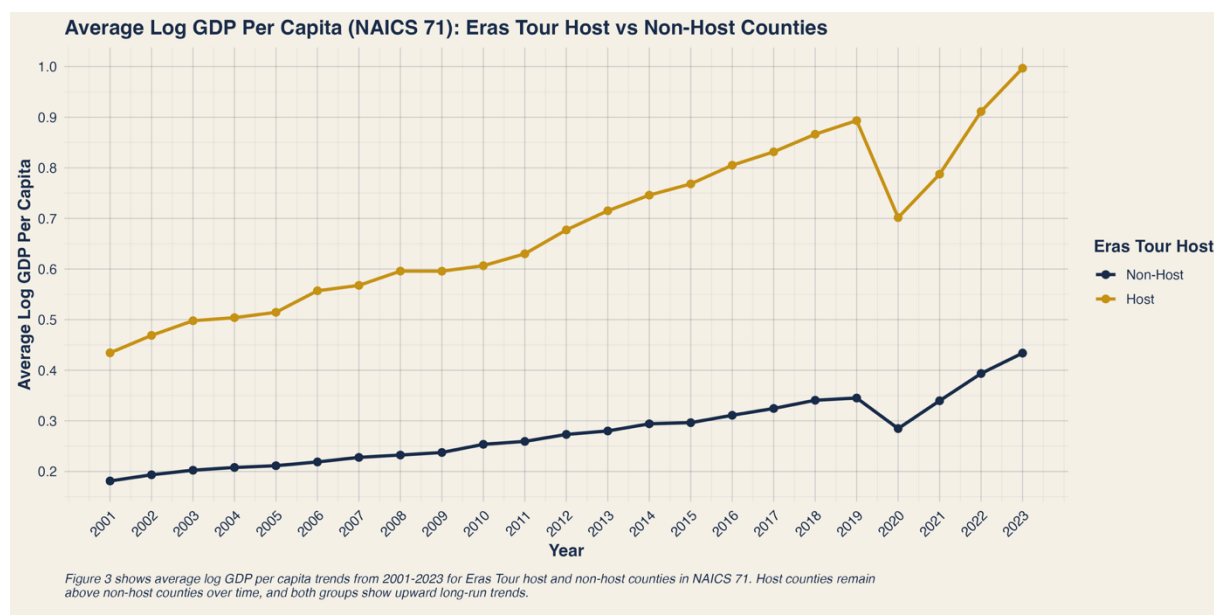


Table 1

Regression Results: NAICS 11 - Agriculture, Forestry, Fishing, and Hunting					
	<i>Dependent variable:</i>				
	Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	-0.2824*** (0.1029)	-0.1251 (0.0979)	-0.0980 (0.0984)	-0.0950 (0.0989)	-0.0157 (0.0324)
Less than HS Share		0.0024 (0.0097)	0.0202 (0.0130)	0.0216 (0.0137)	-0.0083* (0.0045)
HS Only Share		0.0221*** (0.0056)	0.0238*** (0.0057)	0.0242*** (0.0058)	-0.0020 (0.0019)
Some College Share		0.0096 (0.0080)	0.0105 (0.0080)	0.0110 (0.0082)	0.0009 (0.0027)
Bachelors or Higher Share		-0.0044 (0.0045)	-0.0038 (0.0045)	-0.0032 (0.0048)	-0.0025 (0.0016)
Poverty Rate (2021)			-0.0154** (0.0075)	-0.0155** (0.0075)	0.0038 (0.0025)
Net Migration (2023)				0.000001 (0.000003)	-0.000000 (0.000001)
Log GDP Per Capita (2022)					0.8842*** (0.0150)
Constant	0.3257*** (0.0223)	-0.2217 (0.3040)	-0.2330 (0.3029)	-0.2721 (0.3263)	0.1099 (0.1069)
Observations	424	424	424	424	424
R ²	0.0175	0.1749	0.1832	0.1834	0.9129
Adjusted R ²	0.0152	0.1650	0.1715	0.1697	0.9112
Residual Std. Error	0.4491 (df = 422)	0.4135 (df = 418)	0.4119 (df = 417)	0.4124 (df = 416)	0.1349 (df = 415)
F Statistic	7.5350*** (df = 1; 422)	17.7197*** (df = 5; 418)	15.5893*** (df = 6; 417)	13.3487*** (df = 7; 416)	543.4167*** (df = 8; 415)
<i>Note:</i>				*p<0.1; **p<0.05; ***p<0.01	

Table 2

Regression Results: NAICS 54 - Professional, Scientific, and Technical Services					
	Dependent variable:				
	Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	1.1012*** (0.1743)	0.5056*** (0.0908)	0.4386*** (0.0884)	0.4516*** (0.0885)	-0.0002 (0.0098)
Less than HS Share		0.0596*** (0.0099)	0.0133 (0.0128)	0.0211 (0.0136)	-0.0031** (0.0015)
HS Only Share		-0.0069 (0.0057)	-0.0117** (0.0056)	-0.0096* (0.0057)	-0.0005 (0.0006)
Some College Share		-0.0100 (0.0081)	-0.0133* (0.0079)	-0.0106 (0.0080)	0.000001 (0.0009)
Bachelors or Higher Share		0.0767*** (0.0044)	0.0747*** (0.0043)	0.0776*** (0.0046)	-0.0002 (0.0007)
Poverty Rate (2021)			0.0392*** (0.0073)	0.0387*** (0.0073)	0.0007 (0.0008)
Net Migration (2023)				0.000005* (0.000003)	0.000000 (0.000000)
Log GDP Per Capita (2022)					1.0030*** (0.0057)
Constant	1.4187*** (0.0407)	-0.3721 (0.3056)	-0.3010 (0.2947)	-0.5054 (0.3191)	0.0836** (0.0350)
Observations	367	367	367	367	354
R ²	0.0986	0.7725	0.7894	0.7910	0.9977
Adjusted R ²	0.0961	0.7693	0.7859	0.7869	0.9976
Residual Std. Error	0.7579 (df = 365)	0.3829 (df = 361)	0.3689 (df = 360)	0.3680 (df = 359)	0.0391 (df = 345)
F Statistic	39.9160*** (df = 1; 365)	245.1464*** (df = 5; 361)	224.8805*** (df = 6; 360)	194.0667*** (df = 7; 359)	18,369.4700*** (df = 8; 345)
Note:				* p<0.1; ** p<0.05; *** p<0.01	

Table 3

Regression Results: NAICS 71 - Arts, Entertainment, and Recreation

	<i>Dependent variable:</i>				
	Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	0.5629*** (0.0917)	0.3276*** (0.0762)	0.2993*** (0.0763)	0.2973*** (0.0767)	0.0162** (0.0081)
Less than HS Share		0.0389*** (0.0079)	0.0196* (0.0105)	0.0185* (0.0112)	-0.0005 (0.0012)
HS Only Share		0.0079* (0.0046)	0.0058 (0.0046)	0.0055 (0.0048)	-0.0010* (0.0005)
Some College Share		0.0187*** (0.0066)	0.0179*** (0.0066)	0.0176*** (0.0067)	0.0010 (0.0007)
Bachelors or Higher Share		0.0390*** (0.0036)	0.0383*** (0.0036)	0.0379*** (0.0038)	0.0008* (0.0004)
Poverty Rate (2021)			0.0165*** (0.0061)	0.0166*** (0.0061)	0.0021*** (0.0006)
Net Migration (2023)				-0.000001 (0.000002)	0.000000 (0.000000)
Log GDP Per Capita (2022)					1.0385*** (0.0055)
Constant	0.4339*** (0.0206)	-1.1940*** (0.2467)	-1.1761*** (0.2448)	-1.1476*** (0.2642)	-0.0147 (0.0282)
Observations	395	395	395	395	391
R ²	0.0875	0.4148	0.4258	0.4259	0.9938
Adjusted R ²	0.0852	0.4072	0.4169	0.4155	0.9937
Residual Std. Error	0.3996 (df = 393)	0.3217 (df = 389)	0.3190 (df = 388)	0.3194 (df = 387)	0.0333 (df = 382)
F Statistic	37.6810*** (df = 1; 393)	55.1350*** (df = 5; 389)	47.9493*** (df = 6; 388)	41.0142*** (df = 7; 387)	7,654.6130*** (df = 8; 382)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Table 4

Regression Results: NAICS 72 - Accommodation and Food Services					
	Dependent variable:				
	Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	0.5179*** (0.0941)	0.2369*** (0.0716)	0.1874*** (0.0700)	0.2024*** (0.0699)	0.0064 (0.0060)
Less than HS Share		0.0458*** (0.0075)	0.0110 (0.0099)	0.0194* (0.0105)	-0.0014 (0.0009)
HS Only Share		-0.0094** (0.0043)	-0.0134*** (0.0043)	-0.0110** (0.0043)	-0.0004 (0.0004)
Some College Share		0.0270*** (0.0062)	0.0253*** (0.0060)	0.0281*** (0.0061)	-0.0011** (0.0005)
Bachelors or Higher Share		0.0368*** (0.0034)	0.0353*** (0.0033)	0.0384*** (0.0035)	-0.000005 (0.0003)
Poverty Rate (2021)			0.0291*** (0.0056)	0.0285*** (0.0056)	0.0002 (0.0005)
Net Migration (2023)				0.00001** (0.000002)	-0.000000 (0.000000)
Log GDP Per Capita (2022)					1.0284*** (0.0045)
Constant	1.0285*** (0.0212)	-0.4212* (0.2335)	-0.3716 (0.2263)	-0.5915** (0.2433)	0.0886*** (0.0211)
Observations	396	396	396	396	392
R ²	0.0713	0.5015	0.5339	0.5406	0.9966
Adjusted R ²	0.0690	0.4952	0.5267	0.5323	0.9965
Residual Std. Error	0.4102 (df = 394)	0.3021 (df = 390)	0.2925 (df = 389)	0.2907 (df = 388)	0.0249 (df = 383)
F Statistic	30.2656*** (df = 1; 394)	78.4834*** (df = 5; 390)	74.2727*** (df = 6; 389)	65.2173*** (df = 7; 388)	14,035.1700*** (df = 8; 383)

Note:

* p<0.1; ** p<0.05; *** p<0.01

